

Test #4 of Erlang

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1. Write a function `filter/1` such that, if

$$\text{filter}(P) \xrightarrow{+} \{R, S\},$$

then R is the list of the non-list items of P and S is the list of the list in P , in the same order as in P . For example,

$$\text{filter}([a, 1, [], 2, [b, [3]]]) \xrightarrow{+} \{[a, 1, 2], [[], [b, [3]]]\}.$$

$$\text{filter}([a, 1, 2]) \xrightarrow{+} \{[a, 1, 2], []\}.$$

$$\text{filter}([]) \xrightarrow{+} \{[], []\}.$$

2. Strings in Erlang are like in Java. For example, "This is a string". Concatenation of strings is performed by the `++` operator, as in Java; for instance, "This is a" ++ " fruit" evaluates in "This is a fruit". Write a function `punc/1` (*punctuates*) such that, if

$$\text{punc}(P) \xrightarrow{+} S,$$

then S is the string made of the concatenation of the strings in the list P , separated by commas, except the last two strings, which require "and". The resulting string is terminated by a period as well, like in English. For example,

$$\text{punc}(["One", "two", "three", "four"]) \xrightarrow{+} \text{"One, two, three and four."}$$

If there is only one string in P , only the terminal period is required, for instance:

$$\text{punc}(["One"]) \xrightarrow{+} \text{"One."}$$

If P is empty, the result is the empty string "" (no period inside).